Bif401 obj & sub past papers for final term

1. **Seven Steps to Homology Modelling?**

Template recognition and initial alignment • Alignment correction • Backbone generation • Loop modeling
• Side-chain modeling • Model optimization • Model validation

2. **Factors That Participate In Folding Protein?**

The external factors involved in protein denaturation or disruption of the native state include temperature, external fields (electric, magnetic), molecular crowding, and even the limitation of space, which can have a big influence on the folding of proteins.

3. **Role of Amino Acid?**

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it. A large proportion of our cells, muscles and tissue is made up of amino acids, meaning they carry out many important bodily functions, such as giving cells their structure.

4. **Role of Global Protein?**

Global protein function prediction from protein-protein interaction networks. ... The availability of entire genome sequences and of high-throughput capabilities to determine gene coexpression patterns has shifted the research focus from the study of single proteins or small complexes to that of the entire proteome.

5. **Acidic and Basic Amino Acid?**

There are three amino acids that have basic side chains at neutral pH. These are arginine (Arg), lysine (Lys), and histidine (His). Their side chains contain nitrogen and resemble ammonia, which is a base. Their pKa's are high enough that they tend to bind protons, gaining a positive charge in the process.

6. **Points of Bottom up Proteomic?**

The major alternative workflow used in high-throughput proteomics is called top-down proteomics and does not use proteolytic digestion. Essentially, bottom-up proteomics is a means of determining the protein make- up of a given sample of cells, tissues, etc.

7. **Varies RNA Secondary Structure?**

Types of RNA Secondary Structures - I • RNA Complementary bases in primary structure bond together to fold RNA into a secondary structure

Types of RNA Secondary Structures - II • RNA 1' sequence (5' – 3') can fold onto itself to make 2' structures.
• Two previous examples of 2' structures include Helix and Hairpin structures.

Types of RNA Secondary Structures - III • RNA can be fold to form helices, bulge loops, and interior loops. An other 2' RNA structure is the Junction or Intersection

8. **Objective Base Method and Clustering Approach?**

Clustering Approach UPGMA WPGMA Neighbor Joining Single Linkage Complete Linkage
Objective based Methods Least Square Distances Maximum Likelihood Maximum Parsimony

9. Shot Gun and Peptide Mass Finger Printing?

Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

10. Need For Chou Fasman Algorithms?

The Chou-Fasman algorithm is simple in principle.The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

11. Substitution and Indels?

Substitution at adjacent nucleotides (primarily substitutions at two adjacent nucleotides, but substitutions at three adjacent nucleotides have been observed.

Indels, being either insertions, or deletions, can be used as genetic markers in natural populations, especially in phylogenetic studies.

12. Advantages and Disadvantages Of Ab Initio Modeling?

Advantages • Ab Initio methods can fold any target sequence using only physical atomic properties • Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages • Ab initio methods are the very difficult to design (energy function) • These methods are slow due to the huge possibilities • An order of 1012 steps are needed to simulate protein folding for medium sized protein structures.

13. PK Value is Less than PH Why?

The lower the pKa, the stronger the acid and the greater its ability to donate protons. pH depends on the concentration of the solution. This is important because it means a weak acid could actually have a lower pH than a diluted strong acid. On the other hand, the pKa value is a constant for each type of molecule.

14. What Is Hydrophobicity?

In chemistry, hydrophobicity is the physical property of a molecule that is seemingly repelled from a mass of water. In contrast, hydrophiles are attracted to water. Hydrophobic molecules tend to be nonpolar and, thus, prefer other neutral molecules and nonpolar solvents.

15. Difference Between Folding Protein and Denature Protein, Role Of Folding Protein, and Forces Act on Protein / What Factors Are Used For Folding Proceeds?

Protein folding is the physical process by which a protein chain acquires its native 3-dimensional structure, a conformation that is usually biologically functional, in an expeditious and reproducible manner.

Denaturation of proteins involves the disruption and possible destruction of both the secondary and tertiary structures. Since denaturation reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process.

Protein folding is a process by which a polypeptide chain folds to become a biologically active protein in its native 3D structure. Protein structure is crucial to its function. Folded proteins are held together by various molecular interactions The amino acid sequence of a protein determines its 3D structure.

The forces involved in protein folding include:

• Electrostatic interactions, • van der Waals interactions, • Hydrogen bonds, • Hydrophobic interactions

16. Prediction of Protein Structure?

Protein structure prediction is the inference of the three-dimensional structure of a protein from its amino acid sequence—that is, the prediction of its folding and its secondary and tertiary structure from its primary structure.

17. How Can We Predict Amino Acid In Alpha Helix And Beta Sheets?

Proteins that convert from alpha helix to beta sheet: implications for folding and disease. The sequence of a protein normally determines which amino acid residues will form alpha helices, and which one beta sheets, to an extent that allows secondary structure prediction to be made with a reasonable reliability.

20. Chou Fasman (Loop Formation after Alpha and Beta Sheet)? / How Loops Are Found After The Finding Of Alpha And Beta Sheets?

Alpha helices and beta strands are connected by these turns and loops Loops and turns generally lie on
the surfaces of proteins so they often participate in interactions between proteins and other molecules. In a loop, there are no regular structures as can be found in helices or beta strands.

21. The Term Expansy Stands For?

Expert Protein Analysis System

Expasy • Developed by Swiss Bioinformatics Institute (SIB) • Website provides access to databases and tools

• Proteomics, Genomics, Phylogeny, Systems biology, Population genetics, transcriptomics etc. **(PPT)**

22. Differentiate Between Bottom Up Proteomics And Top Down Proteomics?

Bottom Up Proteomics: Measures peptide masses produced after precursor protein's enzymatic digestion.

Top Down Proteomics: Measures intact proteins followed by their peptides after fragmentation.

23. How Proteins Sequences Identified?

Predicting from DNA/RNA sequences. In biology, proteins are produced by translation of messenger RNA (mRNA) with the protein sequence deriving from the sequence of codons in the mRNA. .....Predicted protein
sequences are an important resource for protein identification by mass spectrometry.

24. Why Folding Is Important?

Proteins have several layers of structure each of which is important in the process of protein folding. The first most basic level of this structure is the sequence of amino acids themselves. The sequencing is important because it will determine the types of interactions seen in the protein as it is folding.

25. What Is Ab Initio Modelling? Describe The Limitations Of This Model?

Ab initio methods have Anfinsen'sthermodynamic hypothesis at the center • These methods attempt to identify the structure with minimum free energy. • Applicable to any sequence • Not very accurate biologically • Accuracy & applicability are limited by our understanding of the protein folding problem.

Limitation • Computationally expensive • Suitable for proteins with less than 100 residues (PPT)

Limitation: Models can be conceptual, graphical or mathematical as they are used in science. It allows one to break down a concept into simpler terms with a visual component. A limitation of models in science is that they are usually simplified versions of the real situation or concept.

26. Write Down The Following Terms Of PDB (3)

Title (Title of the structure)
Source (Identify which organism structure from) COMPND (Brief detail of the structure)
REVEDAT (The data of last revision)

27. What Happend When Ph Is Less Than The Pk Value?

If the pH is higher than the pKa, then the compound will be deprotonated. A further consideration is the charge on the compound. Acids are neutral when protonated and negatively charged (ionized) when deprotonated. Bases are neutral when deprotonated and positively charged (ionized) when protonated.

28. Define Ionization In Terms Of Proteins?

Any process by which a neutral atom gains or loses electrons, thus acquiring a net charge, as the dissociation of a substance in solution into ions or ion production by the passage of radioactive particles.

29. Mathematical Relation For Representing Phylogenetic Tree?

A phylogenetic tree is a diagram that represents evolutionary relationships among organisms In trees, two
species are more related if they have a more recent common ancestor and less related if they have a less recent common ancestor. Phylogenetic trees can be drawn in various equivalent styles.

30. Define MGF?

MGF – Mascot Generic Format • MGF is a simple human-readable format for MS/MS data developed by Matrix Science • Mascot Search Engine available at http://www.matrixscience.com/search_form_select.html.

31. What Is Difference Stack Energy And ---------- Energy?

The stacking-fault energy (SFE) is a materials property on a very small scale. ... A stacking fault is an interruption of the normal stacking sequence of atomic planes in a close-packed crystal structure.

32. Summary Of Alpha Helix By Chou's Algorithm?

?????????????

33. How Homology Modelling Is Used To Sequence Unknown. Protein Sequence?

Homology modeling is a procedure that generates a previously unknown protein structure by "fitting" its sequence (target) into a known structure (template), given a certain level of sequence homology (at least 30%) between target and template.

34. Chou Fasman Algorithm?

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman.

35. Improvements Of Chou Fasman Algorithm?

The Chou-Fasman algorithm, one of the earliest methods, has been successfully applied to the prediction. ...

We improved Chou-Fasman method in three aspects. (a) Replace the nucleation regions with extreme values of coefficients calculated by the continuous wavelet transform.

36. Limitations Of Chou Fasman?

The Chou–Fasman method has been widely used for predicting protein secondary structure. It is based on knowledge of the potential of amino acid residues to form α-helical or β-sheet regions in proteins. Our main interest in this study was to examine the reliability of these Chou–Fasman parameters.

37. Beta Sheet By Chou Fasman and Alpha Helix By Chou fasman?

The Chou–Fasman method predicts helices and strands in a similar fashion, first searching linearly through the sequence for a "nucleation" region of high helix or strand probability and then extending the region until a subsequent four-residue window carries a probability of less than 1.

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, ...

38. Propensity?

In the statistical analysis of observational data, propensity score matching (PSM) is a statistical matching technique that attempts to estimate the effect of a treatment, policy, or other intervention by accounting for the covariates that predict receiving the treatment.

39. Fragmentation Of Protein?

Protein fragmentation is a common practice in the study of. protein structure and function. 1'2 Specific cleavage of certain. peptide bonds in the primary structure of a protein is essential in.

40. Hydrophobic Amino Acids?

Hydrophobic Amino Acids. Amino acids are grouped according to what their side chains are like. The nine amino acids that have hydrophobic side chains are glycine (Gly), alanine (Ala), valine (Val), leucine (Leu), isoleucine (Ile), proline (Pro), phenylalanine (Phe), methionine (Met), and tryptophan (Trp).

41. Optimal Energy Function In Protein?

Results: For a single protein sequence, the probability of success (i.e. the probability that the folded state is the lowest energy state) is derived. We then maximize the average probability of success for a set of proteins to obtain the optimal potential energy function.

42. Blast And Fasta Tools In Sequence Alignmnedt And Its Differences?

BLAST is the most widely used tool for the local alignment of nucleotide and amino acid sequences. FASTA is a fine similarity searching tool which uses sequence patterns or words. ... The main difference between BLAST and FASTA is in the similarity searching strategies used in each tool.

43. Protein Scoring?

Protein digestibility-corrected amino acid score (PDCAAS) is a method of evaluating the quality of a protein based on both the amino acid requirements of humans and their ability to digest it.

44. Why Is It Difficult To Predict The Structure Of Protein?

It is much more difficult to compare two models that predict a protein whose structure is unknown and determine which one is more accurate." ... Many protein structures are unknown. Groups around the world are all attempting to determine the structures of proteins that are important for current research.

45. Silico Fragments Comparison?

Matching experimental fragments with in silico fragments is the final resort in protein search & identification

46. How Edman Degradation Work?

Edman degradation. Edman degradation, developed by Pehr Edman, is a method of sequencing amino acids in a peptide. In this method, the amino-terminal residue is labeled and cleaved from the peptide without disrupting the peptide bonds between other amino acid residues.

47. Integrative Scoring Scheme?

Three individual scores can be obtained including 1. MW Match Score 2. PST Match Score 3. In vitro & In silico Match Score.

48. Distinguish In Multiple Sequence Alignment And Pairwise Sequence Alignment? Multiple Sequence Alignment:

A multiple sequence alignment is a sequence alignment of three or more biological sequences, generally protein, DNA, or RNA. In many cases, the input set of query sequences are assumed to have an evolutionary relationship by which they share a linkage and are descended from a common ancestor.

Pairwise Sequence Alignment:

Pairwise Sequence Alignment is used to identify regions of similarity that may indicate functional, structural and/or evolutionary relationships between two biological sequences (protein or nucleic acid).

49. What Are Uses Of Mass Spectrometry?

Mass spectrometry (MS) is an analytical technique that ionizes chemical species and sorts the ions based on their mass-to-charge ratio. In simpler terms, a mass spectrum measures the masses within a sample. Mass spectrometry is used in many different fields and is applied to pure samples as well as complex mixtures.

50. Define Tblastx blastn blast?

tblastx (translated
BLAST) blastn
(translated BLAST)
BLAST performs searches by quick alignments on sequences.

51. Scoring In Homology Modelling?

The scoring functions (1) and (2) are sequence-based: score (1) compares the secondary structure prediction for the target sequence with the actual secondary structure assignment of the template protein, while score
(2) evaluates the sequence similarity between the target and template sequence.

52. Why Model Of Protein Structure Is Necessary?

Proteins are chains of amino acids joined together by peptide bonds. Many conformations of this chain are possible due to the rotation of the chain about each Cα atom. It is these conformational changes that are responsible for differences in the three dimensional structure of proteins.

53. How Positive And Negative Energy Relapsed In A Molecule?

The change in enthalpy is negative in exothermic processes, because energy is released from the system into its surroundings. Generally, a positive change in enthalpy is required to break a bond, while a negative change in enthalpy is accompanied by the formation of a bond.

54. Enlist Few Points In Scope Of Bioinformatics?

The scope of a bioinformatics project (or lab) varies widely depending on the balance of math/stats, computational/software/engineering, and biological/molecular training of the researchers involved and the data orsubject matter they deal with particularly DNA sequences, gene expression, protein, epigenetics.

55. Different Secondary Structures Of RNA?

The secondary structures of biological DNA's and RNA's tend to be different: biological DNA mostly exists as fully base paired double helices, while biological RNA is single stranded and often forms complex and intricate base-pairing interactions due to its increased ability to form hydrogen bonds stemming from the.

56. Silico And Vitro?

In silico is an expression meaning "performed on computer or via computer simulation" in reference to biological experiments.

In vitro studies are conducted using components of an organism that have been isolated from their usual biological surroundings, such as microorganisms, cells, or biological molecules.

57. MS1 And MS2 Discuss?

MS1 reports the intact mass of molecules (proteins or peptides) in the sample • Intact mass can be compared with every protein's mass in database to identify the molecule in sample • Incase multiple candidate proteins are reported, **MS2** can be performed MS2 helps measure fragment peptide masses • MS2 data can used to extract peptide sequence tags

58. What Know About Proteomics?

Proteomics generally refers to the large-scale experimental analysis of proteins and proteomes, but often is used specifically to refer to protein purification and mass spectrometry.

59. Top Down Proteomics?

Top-down proteomics is a method of protein identification that uses an ion trapping mass spectrometer to store an isolated protein ion for mass measurement and tandem mass spectrometry analysis. Top-down proteomics is capable of identifying and quantitating unique proteoforms through the analysis of intact proteins.

60. Summary Of Structural Modelling?

Structural equation modeling is a multivariate statistical analysis technique that is used to analyze structural relationships. This technique is the combination of factor analysis and multiple regression analysis, and it is used to analyze the structural relationship between measured variables and latent constructs.

Protein structure prediction?

There are three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling

Chou – Fasman method?

It is a technique for the prediction of secondary structures in proteins i.e.

Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

- For the appearance of each amino acid in each secondary structure type.
- To predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Chou-fasman algorithm (alpha helix):

- For Alpha Helices, 4 contiguous amino acids are required.
- Their Alpha-Helix propensity should be more than 1.0
- Once this propensity falls below 1.0, Alpha-Helix stops.

Chou- Fasman (Loop form after alpha helix and beta sheet)?

After computing the propensity of alpha helices and beta sheets, we need to settle for loops. Let's see how we can find out the loops using Chou Fasman Algorithm.

For any jth residue in sequence, we calculate f (Total) = f(j) f(j+1) f(j+2) f(j+3) (tetra peptide).
If,
f (Total) > 0.000075.
The average value for P (turn) > 1.00 in the tetra peptide.
The averages for the tetra peptide are such P(a-helix) < P(turn) > P(b-sheet), It is a Turn!

Advantage and disadvantage of Ab initio? Advantages:

- Ab Initio methods can fold any target sequence using only physical atomic properties.
- Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages:

- Ab initio methods are the very difficult to design (energy function).
- These methods are slow due to the huge possibilities.

Bioinformatics role?
An interdisciplinary field
New but rapidly developing field
Low requirement on infrastructure and research equipment Vast opportunities for scientific discovery

PAM matrix?
PAM means "Point Accepted Mutations"
Point accepted mutation is a substitution of one amino acid by another such that the protein functions stays conserved.
PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations. PAM matrices are scoring matrices that are useful in computing sequence alignment scores.

CLUSTALW?
Performs alignment in slow/accurate or fast/approximate
Create multiple alignments, optimize existing alignments, profile analysis, and create phylogenetic trees. CLUSTALW can use multiple file formats including EMBL/SwissProt and Pearson (Fasta) etc

Function of Tandem MS?
Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

Steps present in flow chart of homology modeling?
1. Template recognition and initial alignment.
2. Alignment correction.
3. Backbone generation.
4. Loop modeling.
5. Side-chain modeling.
6. Model optimization.
7. Model validation

How homology modeling is used for knowing the sequence of unknown protein sequence?
If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein. So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence

biotechnologists
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technologist)
biotechnologists

ensures that homology modelling will give accurate results. Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

Silico fragment scoring?

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score

How we can get scoring in BUP proteomic?

There are two approaches to perform BUP:

- Peptide Mass Fingerprinting.
- Shotgun Proteomics

ProSight pattern:

Kelleher et al have developed
an Engine i.e. Prosight PTM
ProSight PTM is the state of the art in top down proteomics search.
Using Prosight PTM, post-translational modifications can be accurately identified.

Top down proteomics:

Measures intact proteins followed by their peptides after fragmentation Score in Protein Fragmentation:- If we can:
Measure the mass of fragments using MS.
Calculate the Protein Fragmentation Techniques theoretical mass of the fragments. Then, we can award score on the basis of the similarity of experimental and theoretical mass.

Molecular biology?

It is the branch of biology that deals with the structure and function of the macromolecules (e.g. proteins and nucleic acids) essential to life. Molecular biology is the study of biology at a molecular level.

Protein Fragmentation techniques?

- Electron Capture Dissociation (ECD).
- Electron Transfer Dissociation (ETD).
- Collision Induced Dissociated (CID).

Types of secondary structures of RNA:
I. Single stranded:

- 3'end may fold on to the 5' end.

2. Helices:

- Double stranded RNA helix of stacked base pairs

3. Hairpin loop:

- The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure

4. Bulge Loops:

- Bulges, are formed when a double-stranded region cannot form base pairs perfectly.

5. Interior loop:

- Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

6. Junctions or intersections:

- Junctions include two or more double-stranded regions converging to form a closed structure.
- The unpaired bases appear as a bulge.

Rooted Un Rooted Tree?

Rooted Tree:

- Each node with descendants represents the inferred most recent common ancestor of the descendants.
- The edge lengths in some trees may be interpreted as time estimates.
- Rooted trees can show temporal evolutionary direction.
- Expensive.

Unrooted:

- Only the relatedness of the leaf nodes.
- Do not require the ancestral root to be known or inferred.
- Less expensive.

Uses:

In bioinformatics, such as

- Rooted and unrooted trees can be used to show phylogenetic relationships between sequences.

1) Fasta Was Develop In(Ans1988)
2) In Corey Paulin Modal Of The Protein Each Atom Is Represented By(Solid Sphere)
3) How Many Storages Use For Structural Modling (3)
4) Modeler Software Use For Homology Modling Which Use(Phython File Format,Not Confrm)
5) Mass Of The Based Proteomic Begin With Measure Of(Intact Protein Amino Acid)
6) Protein Sequence Obtained From(Uni Port)
7) During The Formation Of The 5 Nucleotide Hydrogen Bond Energy Release(-12kcal/Mol)
8) Which Method Is Not Use For Making Polygenatic???Msa
9) Rna Dont Have Structer???Quatnaryy
10) Large Numbr Of Sequence Availble??Publically
11) Coding Region Of Dna Is???Exon
12) Blast Use For???Unknown And Known Protein
13) Expasy Develop By???Sib

Variation In Sequence Develop By??? Insertion,Deletion And Substituin
Bond Form Between C And N Terminas??? Hydrogen Bond
X.Y.Z Position Of Protein Are Available By??? Pdb
Technique Use To Measure The Mass Of Protein???Mass Spactrometry
Large Amino Acid??? Tryptophan
Positive Charge Amino Acid??? Lysine And Arginine
Mrna Has??? Single Strand
C.Alpha Atom Are Traced To Recreat??? 3d Structer Of Protein
Bond Form Between Two Atom??? Energy Release
Provide Clue Of Precurser Protein Peptide Sequence??Pst
Amino Acid Requird For Betasheet??? 3-5
Are Formed By Bonding Betweenci And Ni+4n Protein Backbone??? Alpha Halix
Blasta Stand For??? Basic Local Alignment Search Tool
Amino Acid Have Uniwe Trait Which Can Help In??? Protein Folding
Align Comparison Betweendna/Dna Anddna/Protein??? Fasta.
Ratio Use To Calculate Mass Of Protein??? M/Q.
Treading Not Requied For??? Structer Of Entire Backbone.
Needle Man Wounch Man Or Smith Water Man Algorithm Use For??? Local And Global Alignmnt.

Q.1. In mass electrometer chamber protein is deflected according to its ……
a) charge b)velocity c)charge, mass d) volume

in MS chamber mass/charge ratio can be measured by …… of protein molecules.
a) volume b) mass c) charge d) speed

while scoring protein mass score is awarded on the base of ... b/w experimental and theoretical mass.
a) Distance b) closeness c) equality d) all of these

Tandem MS is used to..... of protein fragments.
a) Ionization b) fragmentation c) measure the mass d) sequence

Fragmentation techniques split the molecules of protein into……..
a) Two part b) three part c) four part d) multiple part

Tandem MS is typically called as,
a) MS b) MS1 c) MS2 d) ALL

Dianamic Programing Recombinant Nucleotide By??? Traseback.
Insertion And Deletion Cause??? Gap.
Information Required For Follding Of Amino Acid Is Present In??? Amino Acid.
Mgf Use For??? Proteomic.
Nj Algorith Use To Pridict The??? 2nd Structer Of Rna..

Each fragmentation technique differ in their work on bases of ,
a) Having different splitting sites
b) Having different time duration
c) Having different mass detection
d) all of these

Mass spectrometer use o measure the mass of protein and ……
A) Amino acids b) lipids c) peptides d) DNA

Protien ionization is the.....step in MS based proteomics.
a) first b) second c) third d) fourth

Ionization process involve gaining and removal of …….
a) electron b) proton c) ion d) atom

molecular weight increases by..... Of proton

a) removing b) adding c) transfer d) All of these

We can score each protein hit in away that the better match gets a ……

a) higher score b) lower score c) equal score d) opposite score

ECD gives out "C" and.....ions,

a) B b) Y c) Z d) A

CID gives out "B" and ..... ions,

a) Y b) Z c) B d) C

PST is abbreviation of…..

a) Protein sequence tags b) protein sequence tools c) Peptide sequence tags d) peptide sequence tools

PST is sequence of peptidesproduced during…..

a) MS b) MS1 c) MS2 d) ALL

Fragmentation at consecutive sits lead to a mass difference equal to that of a single

…..

a) peptide b) amino acid c) protein d) DNA

The data base protein which have matching PSTs to invitro protein are known as

……

a) Sample protein b) candidate protein c) data base protein d) PST protein

Experimental peptides also known as ……

a) data base peptide b) observational peptide c) invitro peptide d) sample peptide

Errors in peak matching also considered by…….

a) Sequence b) fragmentation c) ionization d) scoring scheme

Which one is the right sequence of steps to create a composite protein search engine.

a) PST – MW – In silico comparison b) MW – PST – In silico comparison

c) In silico comparison – MW – PST c) MW – in silico comparison – PST

Q. 22 We can´t add these three scoring comparison until we divide them by their……

a) length b) error c) numbers d) distance

Typtic peptides may reach upto an order of …….

a) 200,000 – 30,000 b) 30,00 – 40,00 c) 300,000 – 400,000 d) 400,00 – 500,00

large scale proteomics is a developed form of

a) mass spectrometry b) BUP technique c) TDP technique d) uniprot technique

Mass spectrometry use to measure.....Ratio of ionized protein or peptide.

a) mass/distanceb) charge/distance c) mass/charge d) distance/charge

Raw file in a format are often found in.....That we have to change in an open

format so as to make it readable.

a) Theoretical form b) decimal form c) percentile form d) binary form

mzData was developed by ……

a) mzML b) HUPO-PSI c) RAW format d) MGF

MASCOT is an online ..... Search engine,

a) Bottom up proteomic b) top down proteomic c) both d) none

is a simple human readable format for MS/MS developed by Matrix Science.

a) MASCO b) ABI c) MGF d) RAW format

theoretical vs experimental fragments comparison helps us as the ..... Stage for short listing candidate protein list.

a) 1st b) 2nd c) d) 4rth

To ensure that all fragments of precursor molecules are also charged, we can use,

a) ESI b) RAW c) BOT d) NONE

Monoisotopic mass is the mass of ……..

a) single isotope b) average isotope c) least abundant isotope d) most abundant isotope

Q. 33 Proteins are assembled by ..... in cellular cytoplasm

a.golgi apparatus b.mitochndria **c.ribosomes** d.nucleaus

Q. 34.Each protein is synthesized by polymerization of .....

a.peptide bond **b.aminoacids** c.nucleotide sequance d.all

Q.35.The characteristics are governed by the elemental composition of an amino acid's .....

a.alpha carbon b.H group **c. R group** d.amino group

Q.36 Since ....and.....introduce very little dipole moments in hydrophobic amino acids, these amino acids are non polar

a.H &C b.H & Y c.Y & H d.A & Z

Q. 37 Mostly .....aminoacids found at the surface of folded proteins

a.polar b.nonpolar c.acidic d.basic

Q.38. At pH=7, five amino acids are charged, ... negatively and..... positively

a.3 & 2 b.3 & 3 **c.2 & 3** d.2 & 2

Q. 39.The charge state depends on the acceptance or donation of.....from the R group side chains

a.electron **b.proton** c.neutron d.both a & b

Q. 40. Amino acids have property such as .....

a.charge state b.polarity c.hydrophobicity **d.all**

Q. 41 Tendem MS helps to measure the moleculer weight of ……

a) isotops b) precursor protein c) ionized fragments d) invitro protien

Q. 42.If pH >pK for aminoacids carboxyle side losses the proton and bocome negative which is .....

a.basic b.neutral c.less acidic **d.acidic**

Q. 43. ..... are key factors participating in protein folding

a.solublity b.reactivity **c.both** d.none

Q. 44.The charge state depends on the acceptance or donation of.....from the R group side chains

a.electron **b.proton** c.neutron d.both a & b

Q. 45. Amno acid with the side chain containing organic acids are soluble in ……

a) ethanol b) water c) lipids d) a and c

Q. 46 .....residue has no R group.

a) cystine b) proline c) glycine d) adinine
Q. 47 proline residue..... the flexibility of polypeptide chain.
a) increase b) decrease c) stabilize d) no effect
Q. 48 A peptide unit has a backbone of polar NH and C=O which is highly…….
a) hydrophilic b) hydrophobic c) neutral d) acidic
Q.49 Functional group of protein always attach to its hydrophobic …….
a) surface b) coat region c) core region d) backbone
Q. 50 After polymerization.....are formed.
a) 3D structure b) alpha helix c) beta sheets d) All of these
. 51 The force involve in protein folding ……
a) electrostatic interactions b) van der waals interactions c) hydrogen bonding d)all
Q. 52 .....folded form of protein is thermodynamically more stable than others.
a) Combined b) native c) partially d) a and b
Q. 53 3´ structural can combine with other polypeptide units to form ……
a) secondary structure b) tertiary structure c) quaternary structure d) all
Q. 54 There is a..... variety of protein structure exist.
a) rare b) restrict c) limited d) huge
Q. 55 Folded sub structures are called ……
a) primary structure b) secondary structure c) tertiary structure d) All
Q. 56 The protein super structure are called ……
a) primary structure b) secondary structure c) tertiary structure d) All
Q. 57 Lysine and tryptophan have respectively..... 1-letter codes.
a) L – T b) S – F c) K – W d) N - A
Q. 58 C- terminus is ..... charge.
a) negatively b) positively c) neutral d) none
Q. 59 ..... bonds are the reason of 2' structure formation.
a) covalent b) hydrogen c) ionic d) All
Q. 60 Loops and Coils are very important 2' structures in that they form ..... of proteins.
a) active site b) inactive site c) acidic site d) basic site
Q. 61 ..... Enforce the 3' structure.
a) Covalent bonds b) Hydrogen bonds c) hydrophobic interaction d) all
Q. 62 Proteins with..... chain can form 4' structures.
a) single b) double c) linear d) multiple
Dihedral Angles always form between ……
a) two points b) three points c) four points d) five points
angles are taken up as a result of folding of protein.

a) phi b) psi c) both d) none

.64 ..... can be used to construct the backbone of a protein towards its visualization.
a) N b) C c) C Beta d) C Alpha
Q. 65 Atoms of phosphorus,sulfur, and chlorine are .....in covalent radius.
a) ~1 Å b) ~2 Å c) ~3 Å d) ~4 Å
Q. 66 The angstrom is used to express ……
a) small biological structures b) lengths of chemical bonds,
c) arrangement of atoms in crystals. d) All

Q. 67 Distances between C-Alphas are measured in the unit "Angstrom" by the process of …….
a) adman degradation b) X Ray crystallography c) Blossom matrix d) PAM Matrix

Q. 68 1 A resolution is better than ……..
a) 9 A b) 8 A c) 7A d) 10 A

Q. 67 PDB contains protein……information
a) structure b) size c) distance d) function

Q. 68 PyMOL and Cn3D is a python-script based tool for visualizing the ……
a) protein size b) protein structure c) amino acids size d) a and b

Q.69 BS Models have atoms as colored balls and intermediate bonds as ……..
a) straw b) spindle c) stick d) small line

Q. 70 In CPK diagrams, each atom is represented by a …….
a) dot sign b) Solid Ball c) doted circle d) solid sphere

According to "Anfinsen's thermodynamic hypothesis" protein folds for making a
……..
a) stable structure b) structure having minimum free energy c) structure having maximum free energy d) a and b

Hence, the basic idea of thermodynamic stability is to .....bonding in order to ……. the free energy.
a) minimizing, maximizing b) maximizing, minimizing c) maximizing, stable
d) b and c

To determine the number of each type of interaction within a protein, we need to find its ……
a) type of amino acid b) types of bonds c) inter-atomic distances d) atomic mass

by looking up at the bond/energy table, we can compute the …….
a) overall energy b) molecular mass c) atomic mass d) bond sequence

If two atoms are participating in a covalent bond, their distance is ……
a) ~0.95 A b) ~0.96A c) ~0.97 A d) ~0.98 A

"Helix Formers" are generally.....amino acids (M, A, L…)
a) acidic b) basic c) hydrophilic d) hydrophobic

Alpha Helices are formed by hydrogen bonding (O-H)between …… and ……. atoms in the protein backbone
a) C – H b) H – N c) C – N d) P - N

resides are needed to make a Beta Strand,typically
a) 3 – 4 b) 5 – 10 c) 6 – 7 d) 5 – 6

Loops are the..... of secondary structure
a) 1st b) 2nd c) d) 4rth

what is made by a single beta sheet when it twists and coils upon itself.
a) Beta Barrel b) beta sandwich c) loops d) coils

are flexible & can adopt multiple conformations.

a) coils b) helices c) loops d) All

are apparently disordered regions with random structure and random length.

a) coils b) helices c) loops d) beta sheets

are Non-functional Combinations of 2' structures)

a) loops b) coils c) beta sheets d) motifs

Domains are semi-independent functional structures in a protein having over ……..

a) ~30 residues b) ~40 residues c) ~50 residues d) ~60 residues

is one such system in which proteins are organized into classes, architecture, topology & homology.

a) CATH b) FSSP c) LOCK d) DOLI

By comparing .....can help us evaluate if the proteins do similar function.

a) sequence b) free energy c) structure d) All

For comparing the complete or partial protein structures, the position of ……. can be used.

a) N terminal b) C terminal c) Alpha Carbon d) functional group

Whole protein structures can be compared by calculating the.....difference b/w their Alpha Carbons positions.

a) distance b) root mean squared c) energy d) All

The.....the RMSD, the similar are the proteins.

a) Equal b) higher c) lower d) random

RMSD tells us about the quality of the matches

a) matches b) mismatches c) sequence d) bonding

There are over a million protein sequences but only about .....protein structures.

a) 10,000 b) 100,000 c) 10,0000 d) 1,000

if we have several..... in the sequence, then we can anticipate that a helix may be

formed by them.

a) Proline b) Thiamin c) Alanines d) Adenine

Product of propensity values is computed for .....for each 2' structure.

a) individual propensity b) overall propensity c) greatest propensity d) All

For Alpha Helices.....contiguous amino acids are required.

a) 6 b) 5 c) 8 d) 4

Q. 98 Alpha-Helix propensity should be more than ..... Once this propensity falls, Alpha-Helix stops.

a) 1.0 b) 1.5 c) 2.0 d) 2.5

Q. 99 …… amino acids are needed to start an Alpha Helix and ..... amino acids for Beta Sheet.

a) 3 , 4 b) 4 , 5 c) 5 , 6 d) 4 , 6

Q. 100 Regions where overlapping alpha-helices and betasheets occur are declared helices.

a) loops b) coil c) helix d) helices

In chou fasman algorithm Alpha Helices can be finalized if their propensity is …….

to the propensity for Beta Sheets.

a) equal b) less c) higher d) different

Chou Fasman Algorithm helps us to predict,

a) Alpha Helices b) Beta Sheets c) Turns d) All

Beta sheets can be predicted from …… amino acid sequences

a) primary b) secondary c) tertiary d) quaternary

Q.104Chou Fasman Algorithm is based on statistical occurrence of.....in known structures.

a) protein b) Amino Acids c) nucleotide d) a and b

Q. 105 Functional evaluation of proteins can be performed by……

a) Structure visualization b) , classification c) prediction d) All

Functional evaluation of proteins is important for understanding …….

a) disease b) designing drugs c) treatment d) ALL

X-Ray Crystallography and NMR Spectroscopy are used to find the structures of ……..

a) DNA b) Amino acid c) proteins d) nucleotides

Q.108it is then possible to identify unknown protein structures by just examining the …… protein sequences

a) heterogeneous b) homologous c) multiple d) binary

Q.109In homology modelling, proteins with similar.....are considered.

a) 1' sequences b) 2' sequences c) 3' sequences d) 4' sequences

Q.110Homology modelling is used to predict structures of proteins having high sequence similarity with other proteins with.....!

a) known structures b) unknown sequence c) similar sequence d) a and c

Q111 Protein structure can be predicted by…..... methods:

a). Homology Modelling b). Threading/Fold Recognition c). Ab Initio Modelling d). All

in homo logy modeling,Definition of Template (……..) & Target (.....)

a) known – unknown b) unknown - known c) known – similar d) unknown - similar

One way to handle loops is to take some residues before and after the insertion as ……. residues.

a) sample b) predicted c) anchor d) similar

The backbone of tyrosine strongly prefers .....rotamers & the real side-chain may fit one of them.

a) 1 b)2 c)3 d)4

in homo logy modeling when protein backbone is ready,then next, ……. were modelled and used to bridge gaps.

a) α helices b) β sheets c) coils d) loops

Loop modeling is ……step of Homology modeling.

a) 2nd b) 3rd c) 4th d) 5th

Homology Modelling ..... study conformational changes.

a) can b) cannot c) may be d) some times can

Q.188 We can overcome the limitations of homology modeling by ……

a) Threading b) Bowie Algorithm c) Ab Initio modeling d) a and c

Q. is a software for homology modelling.

a) Homolog b) Modeller c) iTASSER d) a and b

There are ..... stable folds of protein in nature.

a) 30,000 b) 4,000 c) 40,000 d) 5,000

Several tools are available to perform homology modelling in a programmatic or automated way such as…..

a) Swiss Model b) Robetta c) 3D Jigsaw d)all

Fold recognition is also called …….

a) scoring sequencing c) Threading d) modeling

Homology modelling fails to predict quality structures .....is a technique for predicting these structures.

a) Fold recognition b) Swiss Model c) Threading d) a and c

Threading Scoring typically involves using a.....function based on energy of a molecule

a) Z-Score b) C-score c) A-score d) Y-score

cannot be predicted using threading.

a) invtro protein b) primary protein c) 2ndary protein d) Novel proteins

3D-1D Bowie Algorithm Proposed by Bowie et al in ……

A) 1997 B) 1991 C) 1995 D) 1998

3D-1D Identify amino acids based on ……..

a) protein core b) side chain positioning c) solubility d) All

3D-1D methods convert structure and environment information into ……..

a) software b) packets c) profiles d) subunits

Ab initio modeling is Computationally..... Suitable for proteins with less than …… residues

a) cheap – 500 b) expensive – 100 c) cheap – 100 d) expensive - 500

Ab Initio Modelling methods attempt to identify the structure with minimum ……

a) charge ratio b) distance c) molecular weight d) free energy

Energy released during the.....process is computed for predicting structure.

a) Sequencing b) folding c) breakdown d) bonding

Ab initio methods base their predictions on .....models of folding mechanisms

a) Chemical b) statistical c) physical d) all

Global optimization finds the most …….

a) matching sequence b) stable structure c) similar pattern d) a and c

structure not always at the global minimum.

a) Native b) invitro c) insilico d) stable

Ab initio methods are the very ..... to design.

a) easy b) difficult c) suitable d) accessible

In homology modeling first we perform ……. and then ……..

a) Loop modeling - Backbone generation b) Backbone generation - Loop modeling

There are .....Steps of Homology Modelling.

A) 5 b) 6 c) 7 d) 8

Ab initio modeling is Computationally ……. And..... process.

a) easy and expensive b) Expensive and complex c) cheap and easy

d) difficult and cheap

Ab initio methods predict protein structures by folding proteins based on each constituent atom's ……..

a) volume b) charge c) mass d) all

UPGMA is a ..... method.

a) clustering b) non clustering c) both

Homology modelling is performed in cases of .....identity and alignment score • For the "Twilight zone", other strategies are employed

a) high b) low c) equal d) overall

maximum Parsimony etc are ..... methods (not included in this course).

a) clustring b) nonclustering c) mapping d) visualizing

Online tools: ……..

a) mascot b) sequest c) prosight PC d) all

Number of known protein sequences is..... as compared to known proteins structures

a) larger b) much larger c) smaller d) mech smaller